Transporting Root Evidence for Repeated Adsorption Ranking

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***Abstract*—Pressure and composition sweeps repeatedly rank adsorption materials while reusing the same pure-gas curves. Each mixture score requires an implicit equilibrium, and a warm start reduces its cost only after the candidate reaches the numerical solver. Transporting a previously established root enclosure can exclude a candidate before a new integral evaluation. For binary ideal adsorbed solution theory, two partial-pressure ratios give an arithmetic destination enclosure conditional on supported equilibrium. We derive its containment and widening, and integrate it with distinct eligible lower witnesses and a shared static cap index. Solved brackets and fresh negative threshold tests use the same interface; direct-score predicates and an unknown outcome handle numerical uncertainty. A ten-policy study uses 353 NIST curve pairs and 16,010 CRAFTED configuration pairs. On four 64-query CRAFTED streams, paid transport costs 0.456–0.655 times lean cap indexing; a separate matched study gives 0.485–0.623 times on-demand warm solving, with every paired block favorable. Every original one-query CRAFTED prefix loses to LeanCap and Incumbent; none in the separate study beats on-demand warm solving in all blocks. Matched warm and solved-root controls locate the demonstrated benefit in historical root bounds; additional negative-witness retention does not consistently improve paid time. Size and arrival studies expose further losses. A separately timed 0.01–1 bar sweep adds a NIST mixture relation and shows both gains and losses depending on acquisition pressure and k.**

***Index Terms*—scientific query processing, implicit scores, adsorption, repeated top-k queries, numerical reuse**

I. Introduction

Adsorption screening repeatedly asks which stored materials have the highest mixture selectivity at a specified pressure and feed composition. Ideal adsorbed solution theory (IAST) derives each score from pure-gas isotherms through a nonlinear equilibrium [1], [2]. Pressure and composition sweeps keep the candidate curves fixed while changing the equilibrium for every candidate. Public experimental and simulated collections make this a reusable scientific ranking workload [3]–[5].

The expensive step is obtaining enough information about those implicit scores to determine the top-k result. Prefix integration, numerical continuation and warm starts reduce the cost of evaluating a candidate [6]–[8]. Bounds can instead avoid the evaluation [9], [10]. Finite curve support already supplies query-local score bounds with a static order, so a historical method must improve on indexed bounds and warm solving together. A cached point estimate alone does not establish whether a candidate remains below the new rank boundary.

Our starting point is the interval of roots consistent with earlier numerical work. In binary IAST, the two partial-pressure arguments move in opposite directions as the adsorbed fraction changes. Rescaling those arguments transports a source enclosure into a destination enclosure using arithmetic alone. Its upper endpoint can reject a candidate before probing the destination residual. The same operation applies to the wider interval left by a fresh negative threshold test, allowing information about past outsiders to survive alongside solved roots.

Three obligations make this information usable for ranking. First, destination existence is conditional: an old eligible candidate can become unsupported, so only currently eligible, distinct IDs may establish the kth lower threshold. Second, the source interval must contain its root; a rounded threshold sign must not silently become retained evidence. Third, acquisition, retention and bound evaluation must cost less than the probes they save. We meet the first obligation through a shared support guard and static cap index, the second through guarded direct-score tests and source validation, and test the third with fully paid lifecycles.

We derive an arithmetic root-transport rule with containment and width bounds, use those bounds before numerical evaluation in a ranking executor, and evaluate it with controls for cap indexing, warm solving and retention policy. All ten mechanism policies share the numerical kernel. The strongest observed benefit comes from transported solved-root information on the original near/far CRAFTED streams: Fig. 1 shows acquisition being repaid against two fixed controls. Additional negative witnesses do not yield a consistent paid improvement over solved roots alone. Small pools, large requested answers and wide query changes expose the limits of reuse.

A separately timed comparison starts warm-only execution from an empty cache and retains the long-stream CRAFTED benefit. An operating-range study at independently motivated pressure/feed conditions then varies acquisition pressure and adds a NIST CO_2/N_2 relation. It finds both favorable and losing cases: acquisition state and requested answer size remain part of the performance claim.

II. Related Work

a) IAST solvers and reusable numerical state: IAST infers mixture adsorption from pure-component isotherms [1]. Efficient equilibrium solvers exploit its numerical structure [6]; pyIAST supports fitted and interpolated curves, including analytic integration of piecewise-linear (PWL) segments [2]. RUPTURA caches pressure-related state for later solves, and subsequent formulations further restructure equilibrium computation [7], [8]. Our matched warm control reuses solved brackets in the shared backend. Transport exposes a different

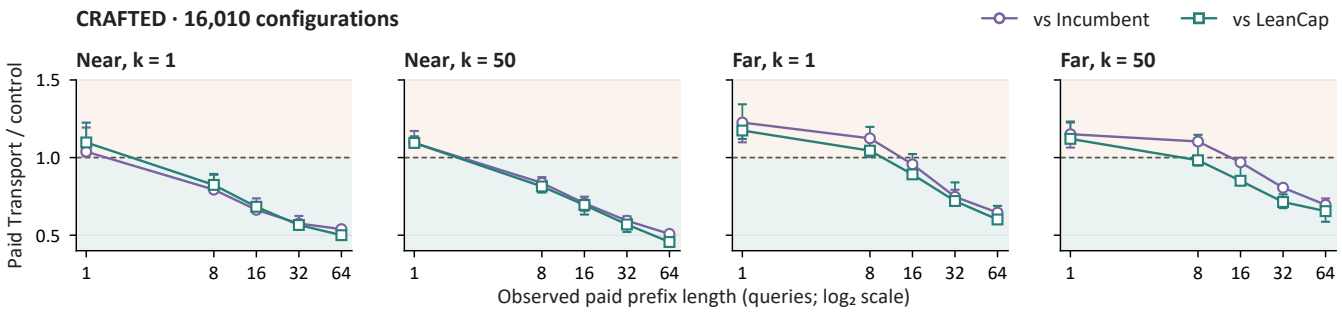


Fig. 1. When paid acquisition becomes useful. For each complete-CRAFTED configuration, curves show Transport/control paid-time ratios at all five measured prefixes against fixed Incumbent and LeanCap controls; points are medians and intervals span three paired timing blocks. Acquisition and teardown are included; connecting lines guide the eye between sampled horizons. Every one-query prefix loses to both controls, while every 64-query comparison wins. The smaller NIST relation includes a losing far/$k = 50$ configuration; Table IV retains all eight configurations.

use of that state: a conditional rank bound available before evaluating the destination residual. The experiments compare implementations within this common kernel; competitiveness with native external solver stacks remains unmeasured.

b) Ranking with expensive scores: Minimal probing avoids expensive score evaluations when available bounds determine the answer [9]. NRA maintains separate lower and upper bounds and stops once selection is established [10]; query-dependent ranking studies score functions whose parameters change across requests [11]. Recent opaque-score ranking learns score distributions and allocates expensive evaluations toward approximate top-k quality [12]. We use bound-based selection for the declared numerical ranking contract. Our bound generator comes from the binary-IAST partial-pressure equations, and accepts both completed root brackets and incomplete evidence from rejected candidates. The kth-lower-bound cut and upper-priority traversal are shared ranking machinery.

c) Retained bounds and deferred computation: Predicate result range caching retains parameter regions over which an expensive predicate's outcome is unchanged [13]. Observed-value envelopes and safe screening reuse information to eliminate computation on related inputs [14], [15]. Lazy exact objects retain certified approximations and defer evaluation until a predicate requires it [16]. Our specialization gives a closed-form, derivative-free finite-step enclosure from source containment, positive curves and partial-pressure ratios. Its exact-real containment proof is separate from the guarded numerical executor, whose library and fallback assumptions are evaluated empirically. The evaluation shares curve prefixes, continuation and the cap index while charging acquisition.

d) Scientific data and computational semantics: Published adsorption data support screening across experimental sources [3], [5], while CRAFTED supplies simulated curves under multiple material configurations [4]. We rank these stored curves under an explicit interpolation and origin convention with finite upper support. The resulting comparisons concern that numerical model; they do not validate physical IAST accuracy or identify improved separation materials.

III. Finite-Curve Ranking Queries

The input relation contains identified pairs of compatible pure-gas curves. Experimental pairs share source, material, temperature and compatible units. Simulated pairs also fix their structural and force-field configuration. A query $z = (P, y, k)$ specifies total pressure $P > 0$, gas-1 feed fraction $0 < y < 1$ and requested answer size $k \geq 1$. Its partial pressures are $f_1 = Py$ and $f_2 = P(1 - y)$.

A. Stored Curves and Supported Equilibria

Each curve has positive loading knots at strictly increasing positive pressures, with final pressure M_i. The stored model interpolates linearly, joins the first knot to the origin, and is undefined above M_i. The origin segment is a modeling convention, not an additional measurement. Loadings need not increase with pressure. Positive loading is sufficient for the spreading-pressure function

$$\Pi_i(p) = \int_0^p \frac{q_i(t)}{t}\, dt, \qquad \Pi_i'(p) = q_i(p)/p > 0 \tag{1}$$

to increase strictly. For a linear segment $q_i(p) = mp + b$, the increment from a to c is $m(c-a) + b\log(c/a)$; pyIAST uses analytic integration of interpolated isotherms [2]. Our common executor stores prefix integrals and uses a segment lookup to evaluate Π_i without scanning all earlier knots.

Write x for the adsorbed mole fraction of gas 1. Binary IAST requires equal spreading pressures [1], [6]:

$$G_f(x) = \Pi_1(f_1/x) - \Pi_2(f_2/(1-x)) = 0, \tag{2}$$

$$D_f = [d_\ell, d_u] = [f_1/M_1,\ 1 - f_2/M_2]. \tag{3}$$

A root must lie in D_f. An empty interval means the requested mixture is unsupported. On a nonempty interval,

$$G_f'(x) = -\frac{q_1(f_1/x)}{x} - \frac{q_2(f_2/(1-x))}{1-x} < 0. \tag{4}$$

Thus a supported root is unique; it exists exactly when the endpoint signs satisfy $G_f(d_\ell) \geq 0$ and $G_f(d_u) \leq 0$. Endpoint equality is eligible.

The common executor precomputes an equivalent support guard. Let $C = \min_i \Pi_i(M_i)$ and $r_i = \Pi_i^{-1}(C)$. The standard spreading-pressure closure is decreasing, so supported existence is equivalent to

$$f_1/r_1 + f_2/r_2 \leq 1. \tag{5}$$

The guard and its construction cost are shared by every compared method. They are a finite-domain feasibility check, not the transport contribution.

B. Scores, Thresholds and Partial Information

For an eligible candidate with root x^*, selectivity is

$$s_y(x^*) = \frac{1-y}{y}\frac{x^*}{1-x^*}. \tag{6}$$

The map is strictly increasing. A score threshold $\tau > 0$ corresponds to

$$x_\tau = \frac{\tau y}{1-y+\tau y}. \tag{7}$$

If $x_\tau > d_u$, any supported score is below τ. If $x_\tau \in D_f$ and $G_f(x_\tau) < 0$, monotonicity proves the same conclusion. A positive residual proves that the root lies to the right and can initialize the subsequent solve. This threshold-to-root continuation is part of the strong baseline.

The same residual can be evaluated directly at a score $t > 0$. Substitution of Eq. (7) gives

$$H_f(t) = \Pi_1(f_1 + f_2/t) - \Pi_2(f_2 + tf_1). \tag{8}$$

Whenever both pure pressures are in support, H_f is strictly decreasing and has the same sign as $G_f(x_t)$. The positive-sum pressure expressions avoid recovering $1 - x_t$ from a rounded fraction. This is an algebraic reformulation of the threshold test, not an additional physical assumption. Numerical sign handling is specified in Section V.

The exact mathematical target is an unordered set of k eligible candidates whose scores dominate all eligible outsiders, or the complete eligible set when fewer than k exist. A selected lower witness must establish eligibility. An outsider's upper bound can be conditional on eligibility. A proved unsupported candidate is excluded by eligibility; an unresolved required solve cannot be assigned a low score and discarded.

The measured service applies the same operational tolerance-shell and candidate-ID rule in every policy, and fails the whole query if required support or root resolution remains unresolved. Section V specifies that contract. Its agreement with the complete numerical reference is an experimental property, separate from the exact-real selection argument.

IV. Transporting Root Information

A genuine source enclosure can come from opposed residual signs or a fresh threshold rejection. Transport moves it to a new query without a spreading-pressure evaluation, conditional on destination existence checked by the shared support guard.

A. An Arithmetic Enclosure

Consider source partial pressures $f = (f_1, f_2)$ with supported root z^* and any genuine enclosure $\ell \leq z^* \leq u$. The endpoints need not belong to D_f. For destination partial pressures f', define $c_i = f_i'/f_i$ and

$$\widehat{\ell} = \min\{c_1\ell,\, 1 - c_2(1-\ell)\}, \tag{9}$$

$$\widehat{u} = \max\{c_1 u,\, 1 - c_2(1-u)\}. \tag{10}$$

Theorem 1 (Conditional root transport)**.** *If the destination query has a supported root x'^*, then*

$$x'^* \in [\widehat{\ell}, \widehat{u}] \cap D_{f'}. \tag{11}$$

The statement requires positive loadings and a valid source root enclosure, but no monotonicity of the loading curves themselves.

Proof. First use the exact source root z^*. If $x'^* < \min\{c_1 z^*, 1 - c_2(1-z^*)\}$, then

$$f_1'/x'^* > f_1/z^*, \qquad f_2'/(1-x'^*) < f_2/(1-z^*).$$

Both roots use valid curve arguments. Strict increase of each Π_i implies $G_{f'}(x'^*) > G_f(z^*) = 0$, a contradiction. Similarly, $x'^* > \max\{c_1 z^*, 1 - c_2(1-z^*)\}$ would imply $G_{f'}(x'^*) < 0$. Both affine maps are increasing. Replacing z^* by its lower enclosure endpoint can only decrease their minimum; replacing it by its upper endpoint can only increase their maximum. This gives Eqs. (9)–(10). Membership in $D_{f'}$ follows from supported existence. □

The proof compares valid roots, so enclosure endpoints may lie outside the physical domain. Genuine source containment remains essential; destination support and independently transported anchors tighten by intersection. The operation uses only ratios, products, minima and maxima, irrespective of knot count. Figure 2 shows how a solved bracket can exclude a destination evaluation while a wider negative witness cannot.

The width makes the cost of a parameter change explicit. Let $m = (\ell + u)/2$, $w = u - \ell$, and $\widehat{w} = \widehat{u} - \widehat{\ell}$ before domain intersection. The two affine maps have positive slopes c_1, c_2, giving

$$\widehat{w} \leq \max(c_1, c_2)w + |c_1 m + c_2(1-m) - 1|\,. \tag{12}$$

Each transformed endpoint is within $\max(c_1, c_2)w/2$ of its midpoint; the absolute term separates the two midpoints. Intersections can only tighten the bound.

For a pressure-only change $c_1 = c_2 = c$, the pre-intersection width is exactly $cw + |c-1|$. For a composition-only change, the additional term is $|(y'-y)(m-y)/(y(1-y))|$. Even an exact source root can therefore produce a loose destination bound. Near extreme feeds, a small parameter change can erase its rank information; large changes can make it uninformative.

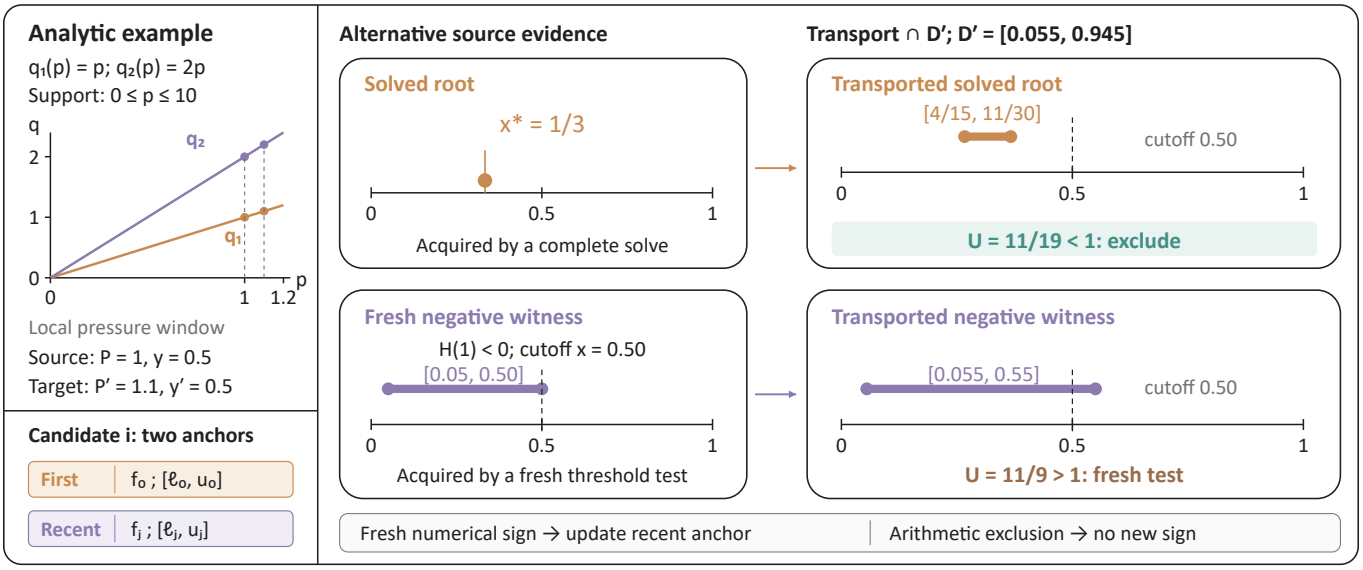


Fig. 2. Two source objects under the same transport rule, shown by an exact analytic example rather than measured curves. For $q_1(p) = p$, $q_2(p) = 2p$ and support $0 \le p \le 10$, pressure changes from $P = 1$ to $P' = 1.1$ at $y = y' = 0.5$ and threshold $\tau = 1$. A paid source solve supplies $x^* = 1/3$; its transported upper score $11/19 < 1$ excludes the destination evaluation. A fresh negative test supplies the wider $[0.05, 0.50]$ interval, whose transported upper $11/9 > 1$ still needs a fresh test. Both alternatives use destination-domain intersection. Arithmetic exclusion creates no new residual sign.

a) Pressure-only exclusion window: For fixed y, valid source upper $0 < u < 1$ and positive effective cut θ, set $x_\theta = \theta y/(1-y+\theta y)$. The raw transported upper $\max\{cu, 1-c(1-u)\}$, $c = P/P_s > 0$, is below x_θ exactly when

$$\frac{1-x_\theta}{1-u} < c < \frac{x_\theta}{u}. \tag{13}$$

Both affine terms must be below x_θ, which gives these inequalities. The window is nonempty iff $u < x_\theta$. For $\gamma = x_\theta - u > 0$, its decrease/increase extents are $\gamma/(1-u)$ and γ/u, with ratio $u/(1-u)$. This is a pointwise consequence for fixed source/cut, before domain/cap intersection, which may exclude outside this window; it is not a trajectory or paid-time guarantee.

B. Information from a Rejected Candidate

A complete root solve supplies a tight source bracket. A threshold rejection supplies a cheaper one. Suppose the source query is supported and an in-domain fresh test proves $G_f(x_\tau) < 0$. Since the root is at least d_ℓ and lies strictly below x_τ,

$$[d_\ell, x_\tau] \tag{14}$$

is a valid source enclosure. It can use the same transport formula on the next query. A candidate eliminated by arithmetic alone supplies no new residual sign; that query cannot be recorded as though a fresh threshold test had occurred.

After acquisition, new solved brackets concentrate on the active frontier. Negative witnesses refresh information about rejected candidates with wider intervals that may still avoid a later integral probe.

The two-slot cache keeps the first acquired anchor and one recent anchor per ID; an unseeded first anchor can be a negative witness. Compatible same-query facts intersect, and a new query replaces the recent slot. Cache size, acquisition and replacement rules are fixed across applicable comparisons.

V. Selection and Numerical Execution

Transport supplies bounds to standard interval top-k selection [10]. Eligibility, distinct witnesses and paid acquisition determine how the executor can use them.

A. Distinct Lower Witnesses and Upper-Bound Scheduling

Mapping a candidate's conditional fraction interval $[\ell_i, u_i]$ through Eq. (6) gives score bounds $[L_i, U_i]$. Let τ_0 be the kth largest established lower bound from distinct, currently eligible IDs, if k exist. Multiple anchors tighten one ID and count once. At least k scores are then at least τ_0, so $U_i < \tau_0$ excludes candidate i from the exact top-k set.

The active threshold is the larger of τ_0 and the minimum lower bound among k solved incumbents. Candidates that survive their bound receive a fresh residual test or root solve. The policy retains resulting brackets or fresh negative witnesses when enabled. The heap control bounds every eligible candidate and visits the largest upper first. The indexed executor below establishes fewer lower witnesses. Successful queries with fewer than k eligible candidates return them all.

A supported equilibrium has spreading pressure at most C. For $R_i(\psi) = \Pi_i^{-1}(\psi)$, $R_i(\psi) \le r_i$ gives the shared query-local caps

$$x^* \ge f_1/r_1, \qquad x^* \le 1 - f_2/r_2. \tag{15}$$

B. A Shared Static Index for Cap Bounds

Applying Eq. (6) to these caps yields

$$U_i^{\mathrm{cap}} = \frac{r_{i2} - f_2}{f_1}, \qquad L_i^{\mathrm{cap}} = \frac{f_2}{r_{i1} - f_1}. \tag{16}$$

Eligible candidates have positive denominators. Thus decreasing r_{i2} orders uppers and increasing r_{i1} orders eligible lowers independently of the query. Both indexes use shared support metadata and are constructed once at paid cost.

The procedure below establishes eligible lower witnesses, visits previous result IDs, then scans the static upper order. A lazy support cache avoids repeated checks. Conditional suffix exclusion requires no individual eligibility decision; lower witnesses and returned answers do. Transport tightens only visited IDs. Cap-only execution shares the orders, incumbents and lower-seed IDs. Warm-only execution shares the historical transform and solve-initialization rule, but uses cap-only lower seeds and arithmetic rejection. Heap–Transport separately materializes all transported intervals.

Numerically, the upper order sorts ascending lower bounds on $1/r_{i2}$; the lower order sorts descending lower bounds on $1/r_{i1}$. The upper coefficient also supplies the outward suffix bound. Candidate-specific physical clipping could destroy that order and is not applied to suffix bounds. Lower seeds require eligibility but need not be the globally best k lower bounds.

Indexed query procedure. $q = (P, y, k)$, $k > 0$; cuts use Eq. (17).

1) Find k eligible distinct lower witnesses, tightening caps with history when enabled. Initialize τ from their minimum, or zero if fewer exist.
2) Visit previous results, then static upper order; skip duplicates. Stop at a static cap upper below the cut.
3) Check support: skip unsupported IDs; abort on unresolved support. Form cap/history bounds for each visited ID.
4) Reject below-cut uppers without creating a source witness.
5) At a positive cut, test the fresh residual: negative rejects and optionally retains a witness; unknown clears threshold hints; positive may supply continuation.
6) Solve with validated initialization. On success, retain the bracket when enabled and update incumbents/τ; on unresolved work, abort the query.
7) Form the shell, refine strict-winner order, then return.

C. Numerical Predicates and Query Outcomes

All arms share C++17 PWL curves, prefixes and numerical policy. Near $y = 1$, rounding x_τ can corrupt its complement. Equation (8) instead forms pure pressures as positive sums directly from the score. Strictly negative residual bounds reject; strictly positive bounds permit continuation. Separated pure-pressure support comparisons can also decide the test. Zero-containing bounds, uncertain domain comparisons or nonfinite expressions return *unknown*, clear threshold-derived witnesses/lower-endpoint hints, and proceed to solving. Separately validated historical initialization remains available.

At requested score tolerance 10^{-12}, successful execution forms a raw top-k heap from solved candidates. For its smallest point score t, set $m(t) = \max(10^{-10}, 10^{-8}|t|)$. Scores above $t+m(t)$ are strict winners; $[t-m(t), t+m(t)]$ is the boundary shell. Keep all strict winners, then fill remaining places by ascending original ID within the shell. Strict winners precede shell members and are score-ordered, with ID breaking exact ties. Adjacent strict winners whose intervals overlap or scores differ by at most $32\,10^{-12}\max(1, |s_i|, |s_j|)$ receive a cold solve if not already canonical. This stabilizes their order without changing shell membership. The same rule orders an under-k eligible set.

For long-double machine epsilon $\varepsilon_{\mathrm{mach}}$, reduce each initial/incumbent lower threshold by $1-1024\varepsilon_{\mathrm{mach}}$ before taking their maximum τ. Arithmetic and fresh-predicate rejection use

$$\tau_- = \tau - 1.001\max(10^{-10}, 10^{-8}|\tau|). \tag{17}$$

Arithmetic exclusion requires $U_i < \tau_-$. Nonpositive cuts retain the candidate for solving. With exact scores and valid bounds, the true shell is preserved: eligible distinct witnesses give $\tau \le t^*$, the exact kth score, and $c(u) = u - 1.001m(u)$ increases for positive u. Hence $c(\tau) \le c(t^*) < t^* - m(t^*)$. Numerical shell membership additionally depends on computed point scores; execution-plan agreement is empirical. A fresh negative supplies an outward fraction upper for its witness. A positive supplies a continuation lower only when interior. Rounded zero supplies no sign.

Adjacent long-double widening covers basic interval operations, including prefixes, coefficients, pressures and coordinate conversions. Each `log1p` is widened by four adjacent values per direction: a library-error assumption, not a verified transcendental bound. Cap inverse estimates propose brackets; bounded integral signs validate endpoints, with uncertain endpoints reverting to zero or maximum pressure. Anchors retain original binary64 P, y; ratios use outward arithmetic. Invalid or empty proposed intersections leave the retained interval unchanged.

Seed–Warm, Roots–Only and Transport pass the same kind of transported interval to the common solver. On its ordinary fast path, the solver intersects this proposal with physical support and accepts it only when destination residual point signs oppose. Failure restores the physical interval, without immediately invoking binary128. A decisive current positive predicate may tighten the lower endpoint and supply continuation. Otherwise the initial point uses the common Henry-slope estimate, replaced by the midpoint if nonfinite or outside the open bracket. No historical point estimate is stored. Initialization rules match; realized histories and thresholds need not.

Newton iteration uses a bisection every eighth step. Final fast brackets require bounded opposed endpoint signs and the score-width target; otherwise the common binary128 selectivity-coordinate solver uses Eq. (8). Its integral budgets are $E_i = 4096\,2^{-112} n_i \mathcal{M}_i$, where n_i is the segment factor and $\mathcal{M}_i$ accumulates absolute intermediate magnitudes. The difference budget is $E_1 + E_2 + 4096\,2^{-112}(|v_1| + |v_2|)$ for integral values v_i. These arithmetic assumptions are implemented in `Curve::precise_pi` and `solve_precise` in the companion's `src/iast_engine.cpp`. An ambiguous residual requires an opposed-sign neighborhood of the requested width.

Otherwise, or after a nonconverged 512-iteration bracket, the solver throws; successful source/score endpoints are converted outward.

Uncertain cap support is reevaluated from original curve/query coordinates using raw binary128 endpoint signs. An unresolved support endpoint also throws. Any such exception aborts the whole query: native return -1, Python `RuntimeError`, no partial ranking. Output arrays are written only after selection; earlier cache updates are not rolled back. Numerical failure is distinct from unsupported equilibrium.

The assurance has three levels: conditional real containment/selection proofs; floating predicates relying on the stated library, support-sign and error-budget assumptions; and empirical execution-plan agreement. The checks in Section VII do not establish all-input floating-point soundness. The current kernel and failed predecessor remain separately versioned, without pooled timings.

D. Specializing the Bound Arithmetic

Positive interval products use lower–lower and upper–upper endpoints. Complementary terms in Eqs. (9)–(10) reverse endpoint directions before outward subtraction. Pressure/feed-ratio intervals use original (P, y) keys and refresh at each public use. Invalid anchors are ignored; unsafe specialization reverts to generic arithmetic.

Direct cap scores avoid the cap–fraction–score round trip near one. The inverse-pressure upper $\overline{r}_i = \mathrm{up}(1/\underline{a}_i)$ uses the static reciprocal coefficient, preserving suffix order at ties. Invalid denominators retain zero lower or infinite upper. Construction and entry points require IEEE binary `Real` and round-to-nearest; compiler flags exclude fast-math. The same-binary generic control retains general `Real`-bound operations under the same spreading-pressure and binary128 assumptions.

E. Costs That Compete with Saved Probes

With n candidates and at most two anchors each, the heap policy pays linear transport/support and heap construction, then $O(a \log n)$ heap work for a active candidates. Static indexing costs $O(n \log n)$ once. Scanning s lower-seed positions, visiting v candidates and maintaining k incumbents after b solves costs $O(s + v + k + b \log(k+1))$, besides numerical work. Retained indexes/caches/anchors use $O(n)$ memory. A query may scan every candidate; there is no sublinear worst-case guarantee.

Each residual uses two spreading-pressure queries with prefix lookup and local evaluation. Arithmetic exclusion avoids both; warm solving reduces subsequent root work. With acquisition/setup A, changed-query costs C_j and teardown D,

$$T(m) = A + \sum_{j=1}^{m} C_j + D. \tag{18}$$

For extra acquisition $\Delta A > 0$ and average query saving $\delta > 0$, crossover requires $m\delta > \Delta A$ after teardown. Observed prefixes test this tradeoff without assuming free acquisition or constant savings.

VI. Evaluation Design

The evaluation tests paid acquisition, information beyond warm solving, and limits under query changes. The original mechanism, size/k and arrival studies use fixed workloads and controls; a separately timed practical study adds on-demand warm acquisition and a published operating range. No cell selects its implementation retrospectively.

A. Relations and Three Workload Studies

The NIST relation is the largest preserved compatible CO_2/CH_4 pool at 298 K, with all 353 curve pairs. The CRAFTED relation contains all 16,010 CO_2/N_2 configuration pairs in its 298 K pool. CRAFTED records are simulated adsorption curves [4], [17]; configurations can share materials and are not independent material samples. NIST records are literature-derived adsorption isotherms from the official mirror [3]. Both relations use the positive-PWL, linear-origin, finite-upper-support convention. Source attribution, selection and conversion records accompany the inputs.

NIST pairing uses publication DOI, material hash and temperature. Among experimental/unspecified duplicates, choose the most retained knots, then source ID. CRAFTED 2.0.1 pairs match material, charge model, force field and temperature. Curves need at least six distinct positive-pressure/positive-loading knots; inconsistent duplicate pressures and pressure reversals are excluded. Units are bar and mmol/g; CRAFTED Pa are converted. These rules precede timing.

The acquisition state is $P_0 = 0.3$ bar and $y_0 = 0.15$. The *mechanism study* uses both complete relations, $k \in \{1, 50\}$ and 64 changed states:

$$P_j = 0.3 \exp\left(a_P \sin\frac{2\pi j}{65} + dj\right), \tag{19}$$

$$\mathrm{logit}(y_j) = \mathrm{logit}(0.15) + a_Y\left(\cos\frac{2\pi j}{65} - 1\right) + 0.7dj. \tag{20}$$

The near regime uses $(a_P, a_Y, d) = (0.04, 0.06, 0.0001)$; the far regime uses $(1.4, 1.7, 0.001)$. All ten arms below are measured at paid prefixes 1, 8, 16, 32 and 64. Both parameters change, preventing an unchanged-result cache from answering the stream.

The *size/k study* uses CRAFTED subsets $n \in \{256, 1024, 4096, 16010\}$ and $k \in \{1, 10, 50, 200\}$. One PCG64 permutation (seed 9201701), independent of curves/scores/performance, supplies every nested prefix and its local IDs, including the full-size relation. All 32 size/k/regime cells run the first 32 states of Eqs. (19)–(20), measuring prefixes 1, 8, 16 and 32. This is dependent within-source variation.

The *arrival study* uses both complete relations, $k \in \{1, 50\}$, two scales and three independently specified random seeds, 31001–31003. For each seed, PCG64 draws 16 independent standard-normal pairs (z_{j1}, z_{j2}) and sets

$$P_j = 0.3 \exp(a_P z_{j1}), \tag{21}$$

$$\mathrm{logit}(y_j) = \mathrm{logit}(0.15) + a_Y z_{j2}. \tag{22}$$

Small/wide arrivals use $(a_P, a_Y) = (0.04, 0.06)$ and $(1.4, 1.7)$, respectively. Identical draws are reused across scales, relations and k, without clipping, resampling or order selection. The 24 distinct cells report prefixes 1, 8 and 16. Size/k and arrivals compare fixed Incumbent, LeanCap, Roots–Only and Transport, retaining unsupported and under-k states. They change queries over existing relations, not the chemistry population.

B. Matched Policies

Table I lists the ten original mechanism arms and the practical study's on-demand control. Transport is the indexed two-anchor policy; Heap–Transport uses the global-heap policy under the same kernel. LeanCap is the principal unseeded arithmetic baseline: it uses the same static cap index without allocating root-history or solver-cache arrays. Its bounds require no historical acquisition. Cap–Index retains the common allocated engine as a separate mechanism control. Seed–Cap additionally pays the complete acquisition query to separate acquisition cost from history use. All seeded arms solve the acquisition state with $k = n$. Indexed arms initially visit at most its best k IDs, then the previous result IDs on later queries, without scanning the full seed result.

Seed–Warm uses the same historical interval transform, destination validation and root initialization as Roots–Only; its lower seeds and arithmetic rejection use only cap information. Roots–Only differs from Transport by suppressing fresh-negative retention. First–Only reads only the first acquired valid anchor; later writes to the second slot remain charged even though that variant does not read them. These comparisons concern evolving policies: tests and retained witnesses can change later caches, so future states are not held fixed.

C. On-Demand Acquisition and Operating-Range Coverage

A separately timed study compares WarmOnDemand with Transport, Roots–Only, LeanCap, Seed–Warm and Incumbent at all five prefixes. WarmOnDemand uses Seed–Warm's initialization rule but starts without a complete seed, retaining every actually solved root, including candidates displaced from the answer. All six policies first repeat the eight original near/far configurations; their new clocks remain separate.

The operating-range study uses CO_2/N_2, 298 K and $y = 0.15$. Chola et al. report these conditions at 0.01, 0.1 and 1 bar [18]. Before timing, we fix 64 ascending targets $P_j = 10^{-2+2j/63}$ bar, $j = 0, \ldots, 63$, and alternative acquisition pressures 0.01 and 1 bar. Grid, order and acquisition choices are ours, not an observed session or process cycle. Each seeded lifecycle pays for one acquisition state.

Both $k = 1/50$ use CRAFTED's 16,010 pairs and all 291 compatible NIST CO_2/N_2 pairs. Of the latter's 255 material IDs, 128 overlap the original NIST CO_2/CH_4 pool: the new mixture relation is not disjoint material sampling. All targets/candidates remain under the finite-support guard, without extrapolation or dropping under-k states. Two datasets, two acquisition states and two k values give eight configurations, each with the same prefixes and three paired blocks.

D. Paid Costs and Descriptive Repetition

The original measurements use an AMD EPYC 9V74 host under Linux 6.18.55, with each producer pinned to CPU 0. The container has eight CPU equivalents and 8 GiB of memory. The implementation uses Python 3.12.14, NumPy 2.3.5 and G++ 13.3.0, with C++17, `-O3`, `-ffp-contract=off`, `-frounding-math` and `libquadmath`. The practical study uses the same processor, CPU affinity, compiler flags, Python and NumPy versions under Linux 6.18.57, with unchanged kernel and wrapper source bytes. Each study fixes one binary, excludes diagnostics from timing, and keeps its clocks separate. Three timing blocks rotate arm order and reverse it in odd blocks; each arm, cell and block owns a fresh engine. Integral-probe counts record scalar or bounded spreading-pressure API calls. Bounded calls can evaluate two endpoints across knots; counts therefore measure neither transcendental operations nor constant-cost work across versions.

Paid lifecycles include native curves and support metadata, any static-index construction, resets, full-seed acquisition when present, incumbent handling, queries, output conversion and teardown. Common JSON loading, immutable pool preparation, binary compilation/loading, reference checks and logging are outside every timer. Each prefix adds final teardown for its fixed allocation. The artifact's setup field measures native engine construction, excluding index setup and other preparation.

Ratios pair arm/control observations within a block. Medians and observed ranges are descriptive, not population confidence intervals. A paid advantage requires lower time in all three blocks and numerical agreement. Arrival seeds remain distinct workload rows; they do not replace timing blocks.

LeanCap omits anchor and separate solver-cache arrays. Other arms retain the common allocation, even when unused; Cap–Index exposes this cost. Memory reports owned capacities of curve payloads/objects, pair metadata, anchors, solver caches, indexes and engine, with valid anchors counted separately. It excludes transients, allocator overhead, Python buffers and RSS.

E. Diagnostics and Numerical Checks

A separate untimed diagnostic replay per cell/arm must match timed IDs, scores, brackets and original nonclock counters. Origin bits identify solved, negative and same-state mixed anchors. Width means divide cap-before/final-after width sums by valid transport invocations, excluding the near-support bypass; lower-seed and processing calls can count the same ID twice.

An exclusion is *history-needed at the current threshold* if its combined upper is below the cut but its cap upper is not. This is not counterfactual saved work: history also changes thresholds/order. Tail cuts may include unchecked IDs; global support comes from complete references. First/last $k + 1$ reference queries give computed-model gaps, undefined when too few candidates are supported.

Each timed query must match a cold complete solve's ordered IDs and scores within relative 10^{-8} and absolute 10^{-10}

TABLE I

POLICY DEFINITIONS. THE ORIGINAL MECHANISM STUDY USES THE FIRST TEN ARMS; THE PRACTICAL STUDY ALSO USES WARMONDEMAND. EVERY INDEXED ARM SHARES STATIC ORDERS, LOWER-SEED IDS, LAZY ELIGIBILITY, INCUMBENT HANDLING AND THE NUMERICAL BOUNDARY POLICY. A PAID SEED IS A COMPLETE INITIAL SOLVE, CHARGED IN EVERY PREFIX.

Arm	Acquisition	Traversal	Historical information used
Incumbent (I)	none	previous results, then fixed scan	previous result order only
Cap–Heap (C–H)	none	global cap-upper heap	none
Cap–Index (C–I)	none	shared static cap index	none; common history allocation
LeanCap (L)	none	shared static cap index	none; history/cache arrays omitted
Seed–Cap (S–C)	complete seed	shared static cap index	seed incumbent IDs only
Seed–Warm (S–W)	complete seed	shared static cap index	solved brackets for root initialization only
Roots–Only (R)	complete seed	shared static cap index	solved brackets for lower/upper bounds and solving
First–Only (F)	complete seed	shared static cap index	first acquired anchor per ID
Transport (T)	complete seed	shared static cap index	first and recent solved/negative-witness anchors
Heap–Transport (H–T)	complete seed	global transported-upper heap	first and recent solved/negative-witness anchors
WarmOnDemand (D)	none	shared static cap index	actually acquired solved brackets for initialization only

tolerances. This checks paths, not independent arithmetic. Separate checks reconstruct PWL integrals from exact stored binary64 inputs in 80-digit Decimal, and audit raw accounting, reductions and boundary fixtures.

At fixed full-pool audit states, independently recompute the shared reference's initial top-k IDs. Their minimum supported score s sets $c = \max(0, s - 2\max(10^{-10}, 10^{-8}|s|))$. Scan every supported candidate against c and independently solve every positive/ambiguous contender. If fewer than $\min(k, n_{\text{supported}})$ initial IDs remain, solve all supported roots. The initial IDs need not be the winners: any k independent scores establish a permissive cutoff. Scaling uses $k = 200$, arrivals $k = 50$; saved-reference reuse remains separate. Section VII specifies coverage.

VII. RESULTS

The mechanism, size/k and arrival studies contain 15,360, 12,288 and 4,608 timed outputs. All 32,256 expected queries returned successfully and agree with their shared complete references. The two practical-study cohorts each contain another 9,216 timed outputs with complete-reference agreement. All cohorts retain separate clocks.

A. Paid Historical Information

Table II reports every mechanism arm at 64 states. On CRAFTED, Transport/Incumbent median paired ratios are 0.539, 0.508, 0.646 and 0.696 for near $k = 1/50$ and far $k = 1/50$; every block is below one. Ratios to LeanCap are 0.500, 0.456, 0.601 and 0.655, and to Seed–Warm are 0.466, 0.431, 0.536 and 0.536, again favorable in every block. Thus the long-stream benefit survives both a lean static-cap executor and matched warm solving. Comparisons with Heap–Transport are mixed in all four cells, so this does not establish a universally better scheduling policy.

NIST retains an adverse regime. Near $k = 1/50$ and far $k = 1$ beat both Incumbent and LeanCap in every block, but far $k = 50$ loses to Incumbent, 1.391 [1.350,1.845], and is mixed against LeanCap, 1.029 [0.882,1.523]. A larger requested answer and changed state can remove the useful pruning advantage.

Acquisition must also be repaid. At one changed query, all four CRAFTED cells lose to both Incumbent and LeanCap in every block. LeanCap ratios are 1.097, 1.095, 1.174 and 1.121 in the same configuration order. Table IV locates this acquisition crossover for all eight configurations and four fixed controls. On CRAFTED, the first all-block advantage over Incumbent/LeanCap occurs at 8/8 queries for both near cells, 16/32 for far $k = 1$, and 16/16 for far $k = 50$. Within those original comparisons, every observed all-block advantage persists at later measured prefixes; NIST far $k = 50$ never achieves one against any of these controls. These are sampled crossover horizons, with no interpolation of an unmeasured break-even point. Table III exposes construction, seed phase and retained capacity for the fixed examples. LeanCap allocates no anchor payload; the allocated cap control reserves history without valid anchors. These construction and acquisition costs remain included in every paid prefix.

B. On-Demand Warm Solving and Acquisition Choice

Table V repeats the near/far streams against empty-cache WarmOnDemand. At 64 CRAFTED queries, Transport ratios are 0.485, 0.492, 0.584 and 0.623 in near/far $k = 1/50$ order; every block favors Transport and Roots–Only, while all four Seed–Warm comparisons lose. Complete acquisition alone does not explain the benefit. Transport first wins all blocks at 8, 8, 16 and 16 queries. NIST–CH_4 Transport wins all four final comparisons; Roots–Only is mixed at both $k = 50$ cells. Prefix sequences preserve reversals, including a Roots–Only near/$k = 1$ win at 8 becoming mixed at 16.

On the operating grid, CRAFTED acquisition at 0.01 bar gives Transport/WarmOnDemand ratios 1.195 and 1.146 for $k = 1/50$, losing every block at 64 despite both winning at prefix 8. Acquisition at 1 bar gives 0.609 and 0.724, winning every block. Roots–Only has the same final split. On NIST–N_2, both policies win at $k = 1$ and lose at $k = 50$ for both acquisition states. These preselected sources share the target grid: acquisition state, not only horizon, affects paid benefit.

Eligible counts decline from 280 to 90 on NIST–N_2 and 15,948 to 13,242 on CRAFTED, changing acquisition cost

TABLE II

COMPLETE 64-QUERY PAID LIFECYCLES (SECONDS): DESCRIPTIVE MEDIANS OF THREE BLOCKS, INCLUDING PREPARATION, ACQUISITION AND TEARDOWN. COLUMN LABELS FOLLOW TABLE I. RELATIVE TO FIXED INCUMBENT, TEAL ↓ DENOTES FASTER IN ALL THREE BLOCKS, AMBER ∼ MIXED/EQUAL, AND COPPER ↑ SLOWER IN ALL THREE; CLASSIFICATIONS USE UNROUNDED PAIRED TIMES.

Relation	Regime	k	I	C–H	C–I	L	S–C	S–W	R	F	T	H–T
NIST	near	1	0.0640	0.0786$^{\uparrow}$	0.0614$^{\sim}$	0.0560$^{\sim}$	0.0547$^{\downarrow}$	0.0576$^{\downarrow}$	0.0399$^{\downarrow}$	0.0399$^{\downarrow}$	0.0417$^{\downarrow}$	0.0567$^{\downarrow}$
NIST	near	50	0.0946	0.1406$^{\uparrow}$	0.1221$^{\uparrow}$	0.1189$^{\uparrow}$	0.1205$^{\uparrow}$	0.1268$^{\uparrow}$	0.0903$^{\downarrow}$	0.0886$^{\sim}$	0.0890$^{\downarrow}$	0.0881$^{\downarrow}$
NIST	far	1	0.0660	0.0704$^{\sim}$	0.0611$^{\downarrow}$	0.0609$^{\downarrow}$	0.0636$^{\downarrow}$	0.0710$^{\sim}$	0.0472$^{\downarrow}$	0.0505$^{\downarrow}$	0.0511$^{\downarrow}$	0.0600$^{\downarrow}$
NIST	far	50	0.0908	0.1414$^{\uparrow}$	0.1109$^{\uparrow}$	0.1226$^{\uparrow}$	0.1328$^{\uparrow}$	0.1277$^{\uparrow}$	0.1194$^{\uparrow}$	0.1196$^{\uparrow}$	0.1263$^{\uparrow}$	0.1269$^{\uparrow}$
CRAFTED	near	1	4.9496	5.6670$^{\uparrow}$	5.2479$^{\uparrow}$	5.2196$^{\sim}$	5.3186$^{\uparrow}$	5.6033$^{\uparrow}$	2.4741$^{\downarrow}$	2.6367$^{\downarrow}$	2.6127$^{\downarrow}$	2.5468$^{\downarrow}$
CRAFTED	near	50	5.0891	6.1336$^{\uparrow}$	5.3282$^{\sim}$	5.4823$^{\uparrow}$	5.4862$^{\sim}$	5.9882$^{\uparrow}$	2.5255$^{\downarrow}$	2.5605$^{\downarrow}$	2.5477$^{\downarrow}$	2.6762$^{\downarrow}$
CRAFTED	far	1	4.8992	5.9022$^{\uparrow}$	5.1612$^{\sim}$	5.1353$^{\sim}$	5.6349$^{\uparrow}$	5.8535$^{\uparrow}$	3.0149$^{\downarrow}$	3.0625$^{\downarrow}$	3.1525$^{\downarrow}$	3.2343$^{\downarrow}$
CRAFTED	far	50	4.8981	6.4371$^{\uparrow}$	5.7606$^{\uparrow}$	5.7203$^{\uparrow}$	5.7187$^{\uparrow}$	6.7551$^{\uparrow}$	3.4831$^{\downarrow}$	3.6469$^{\downarrow}$	3.6085$^{\downarrow}$	3.6615$^{\downarrow}$

TABLE III

CONSTRUCTION AND RETAINED CAPACITY FOR THE FIXED NEAR/$k = 1/64$-QUERY EXAMPLES. TIMES ARE SEPARATE MEDIANS IN MILLISECONDS. CONSTRUCTION EXCLUDES INDEX SETUP; UNSEEDED SEED-PHASE ENTRIES ARE BOOKKEEPING. OWNED CAPACITY EXCLUDES TRANSIENT, ALLOCATOR AND PYTHON MEMORY AND RSS. ANCHOR BYTES AND VALID SLOTS DISTINGUISH ALLOCATION FROM RETAINED FACTS.

Relation	Policy	Paid	Construct	Seed phase	Destroy	Owned (MiB)	Anchors (KiB)	Valid slots
NIST	LeanCap	55.99	22.48	0.01	0.45	2.516	0.0	0
NIST	Cap–Index	61.36	21.50	0.01	0.46	2.586	55.2	0
NIST	Transport	41.67	21.55	2.58	0.40	2.586	55.2	312
CRAFTED	LeanCap	5219.56	656.08	0.03	13.75	65.268	0.0	0
CRAFTED	Cap–Index	5247.91	624.33	0.03	14.27	68.444	2501.6	0
CRAFTED	Transport	2612.74	636.33	141.11	14.35	68.444	2501.6	15498

TABLE IV

FIRST MEASURED ALL-BLOCK ADVANTAGE, THEN OUTCOMES AT $1, 8, 16, 32, 64$ QUERIES: TEAL W = ALL THREE RATIOS BELOW ONE; COPPER L = ALL ABOVE ONE; AMBER M = MIXED/EQUAL. A DASH MEANS NO OBSERVED ADVANTAGE. EVERY FIRST ADVANTAGE PERSISTS AT LATER MEASURED HORIZONS; NO UNMEASURED BREAK-EVEN TIME IS INTERPOLATED.

Data	Stream	k	Inc	Lean	SeedWarm	RootsOnly
NIST	near	1	16 M M W W W	32 M M M W W	8 M W W W W	– M M M M M
NIST	near	50	32 M M M W W	8 M W W W W	1 W W W W W	8 M W W W W
NIST	far	1	16 L M W W W	64 L L M M W	16 M M W W W	– L L L L L
NIST	far	50	– L L L L L	– L M M M M	– M M M M M	– M M M M L
CRAFTED	near	1	8 L W W W W	8 L W W W W	8 M W W W W	– M M M M M
CRAFTED	near	50	8 L W W W W	8 L W W W W	8 M W W W W	– M M M L M
CRAFTED	far	1	16 L L W W W	32 L L M W W	1 W W W W W	– M L M M L
CRAFTED	far	50	16 L L W W W	16 L M W W W	1 W W W W W	– M M M M M

and available source roots together. Candidate-state support is 72.8%/95.1%. Using Table VI's definition, first/last normalized gaps for $k = 1/50$ are 0.292/0.223 and 0.00311/0.0511 on NIST, versus 0.000777/0.000725 and 0.0166/0.00121 on CRAFTED. Equation (13) explains why pressure distance alone does not determine exclusion. These descriptors neither isolate bound tightness as the paid-time cause nor measure physical separation accuracy.

C. Which Information Is Useful?

Figure 3 distinguishes paid outcomes from query-only decision diagnostics. On CRAFTED near $k = 1$, 949,984 arithmetic exclusions need history at the actual threshold and 704 are cap-sufficient; far $k = 1$ gives 886,979 and 704. These counts do not estimate counterfactual savings. Mean cap/final fraction widths are 0.861/0.035 near and 0.893/0.545 far per valid transport invocation, consistent with the widening in Eq. (12); no width-to-runtime model is fitted. Table VI retains all support and endpoint-gap descriptors.

A post-hoc paired contrast of the same fixed measurements verifies the solved-root contribution directly. At 64 CRAFTED queries, Roots–Only beats Incumbent, LeanCap and Seed–Warm in every block of all four cells; its median ratios to Seed–Warm are 0.431, 0.432, 0.515 and 0.514 in near/far $k = 1/50$ order.

Fresh-negative retention adds information without a consistent paid benefit. CRAFTED Transport/Roots–Only comparisons are mixed for both near cells and far $k = 50$; far $k = 1$ loses every block, 1.070 [1.024,1.105]. In that cell, both policies perform 222 query roots, while Transport uses

TABLE V

SEPARATELY MEASURED PRACTICAL COMPARISONS: ALL 16 CONFIGURATIONS, WITH FIXED WARMONDEMAND (D) AS DENOMINATOR. T: TRANSPORT; R: ROOTS–ONLY; S–W: SEED–WARM. THE THREE RATIOS ARE MEDIAN [MINIMUM, MAXIMUM] PAID TIME AT 64 QUERIES ACROSS THREE PAIRED TECHNICAL BLOCKS, NOT CONFIDENCE INTERVALS. TEAL ↓/W MEANS ALL BLOCKS FASTER; AMBER ∼/M MIXED OR EQUAL; COPPER ↑/L ALL SLOWER. THE FIVE-LETTER SEQUENCES RETAIN EVERY PREFIX 1, 8, 16, 32, 64 IN THAT ORDER; CLASSIFICATIONS USE UNROUNDED VALUES. D HAS NO ACQUISITION QUERY. REMEASURED NEAR/FAR AND OPERATING-RANGE CLOCKS ARE SEPARATE FROM THE ORIGINAL STUDY.

Relation	Regime/P_0 (bar)	k	Paid ratios at 64 queries			Prefix status	
			T/D	R/D	S–W/D	T/D	R/D
Remeasured near/far streams; seeded policies acquire at $P_0 = 0.3$ bar, $y_0 = 0.15$							
NIST–CH_4	near	1	0.617 [0.571,0.782]$^{\downarrow}$	0.631 [0.591,0.857]$^{\downarrow}$	1.388 [0.856,2.034]$^{\sim}$	MMWWW	MWMWW
NIST–CH_4	near	50	0.767 [0.604,0.872]$^{\downarrow}$	0.824 [0.602,1.080]$^{\sim}$	1.178 [0.784,1.497]$^{\sim}$	LMMMW	MMMMM
NIST–CH_4	far	1	0.833 [0.773,0.834]$^{\downarrow}$	0.801 [0.789,0.968]$^{\downarrow}$	1.235 [1.160,1.252]$^{\uparrow}$	LLMWW	LLLMW
NIST–CH_4	far	50	0.961 [0.955,0.961]$^{\downarrow}$	1.083 [0.971,1.097]$^{\sim}$	1.074 [0.992,1.123]$^{\sim}$	MMMWW	LLLMM
CRAFTED	near	1	0.485 [0.464,0.534]$^{\downarrow}$	0.473 [0.455,0.527]$^{\downarrow}$	1.130 [1.121,1.132]$^{\uparrow}$	LWWWW	LWWWW
CRAFTED	near	50	0.492 [0.487,0.501]$^{\downarrow}$	0.486 [0.462,0.501]$^{\downarrow}$	1.136 [1.133,1.157]$^{\uparrow}$	LWWWW	LWWWW
CRAFTED	far	1	0.584 [0.567,0.620]$^{\downarrow}$	0.589 [0.486,0.595]$^{\downarrow}$	1.122 [1.038,1.139]$^{\uparrow}$	MMWWW	MMWWW
CRAFTED	far	50	0.623 [0.603,0.639]$^{\downarrow}$	0.585 [0.554,0.593]$^{\downarrow}$	1.144 [1.063,1.156]$^{\uparrow}$	LMWWW	LMWWW
Operating-range pressure grid; seeded policies acquire at the displayed P_0, $y_0 = 0.15$							
NIST–N_2	0.01	1	0.950 [0.795,0.956]$^{\downarrow}$	0.740 [0.720,0.982]$^{\downarrow}$	1.073 [0.787,1.124]$^{\sim}$	WWWWW	WWWWW
NIST–N_2	0.01	50	1.099 [1.079,1.248]$^{\uparrow}$	1.164 [1.046,1.212]$^{\uparrow}$	1.054 [1.049,1.126]$^{\uparrow}$	MMLLL	LLLLL
NIST–N_2	1	1	0.942 [0.875,0.972]$^{\downarrow}$	0.867 [0.791,0.871]$^{\downarrow}$	1.059 [0.993,1.103]$^{\sim}$	MMMWW	MMWWW
NIST–N_2	1	50	1.103 [1.050,1.120]$^{\uparrow}$	1.103 [1.006,1.208]$^{\uparrow}$	1.002 [0.919,1.025]$^{\sim}$	MMMML	MMMML
CRAFTED	0.01	1	1.195 [1.180,1.197]$^{\uparrow}$	1.072 [1.066,1.083]$^{\uparrow}$	1.111 [1.100,1.150]$^{\uparrow}$	LWMLL	LWMLL
CRAFTED	0.01	50	1.146 [1.130,1.215]$^{\uparrow}$	1.039 [1.016,1.040]$^{\uparrow}$	1.145 [1.079,1.147]$^{\uparrow}$	LWMLL	LWWML
CRAFTED	1	1	0.609 [0.573,0.638]$^{\downarrow}$	0.560 [0.545,0.591]$^{\downarrow}$	1.149 [1.036,1.214]$^{\uparrow}$	LWWWW	LWWWW
CRAFTED	1	50	0.724 [0.672,0.745]$^{\downarrow}$	0.609 [0.607,0.629]$^{\downarrow}$	1.110 [1.105,1.129]$^{\uparrow}$	LMWWW	LMWWW

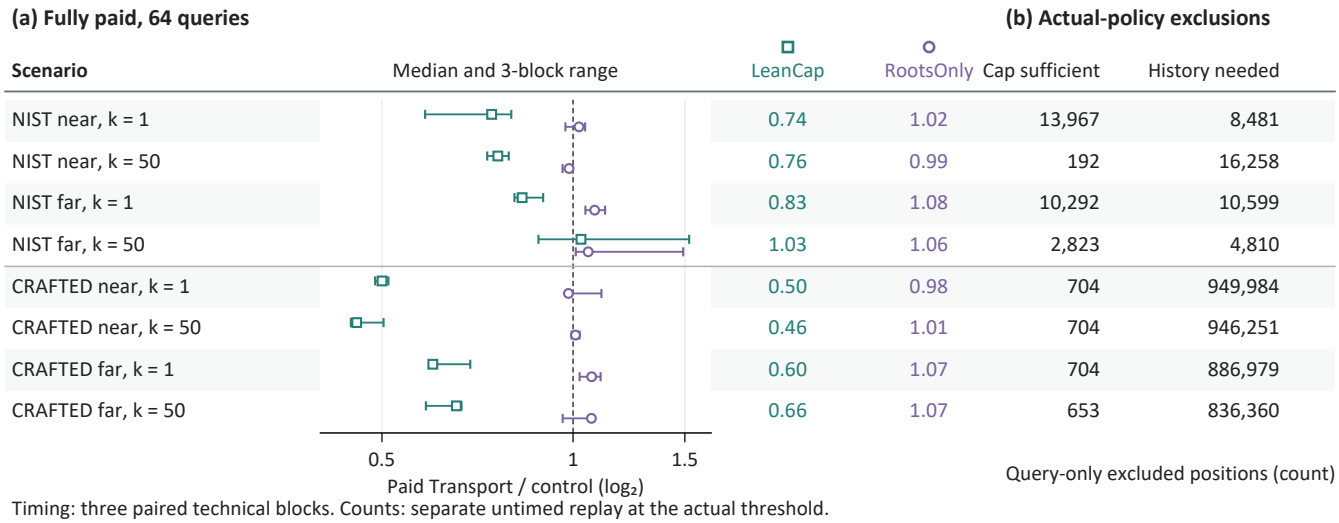


Fig. 3. All eight mechanism configurations at 64 states. Left: paid Transport/control ratios against LeanCap and Roots–Only, with median and minimum/maximum of three paired blocks; color and marker identify the fixed control. Right: exact query-only arithmetic-exclusion counts from the diagnostic replay. History-needed is classified at the actual threshold, not as counterfactual savings. Cap-sufficient includes conditional tail cuts, whose IDs may have unchecked eligibility.

199,082 integral API calls versus 297,708. Lower API-call counts do not establish lower paid time. Transport beats First–Only in both near cells, loses at far $k = 1$ and is mixed at far $k = 50$. The mathematical interface accepts incomplete evidence, but its unconditional retention is not a uniformly profitable policy.

D. Size, Result Count and Arrivals

Figure 4 retains every size/k/regime cell against fixed LeanCap and Incumbent controls. At $n = 16{,}010$, all eight cells beat both in every block over 32 states. Near Incumbent ratios for $k = 1, 10, 50, 200$ are 0.588, 0.572, 0.652 and 0.626; far ratios are 0.770, 0.750, 0.784 and 0.878. At $n = 256$,

TABLE VI
ALL MECHANISM-STUDY SUPPORT AND BOUNDARY-GAP DESCRIPTORS. SUPPORT IS THE PERCENTAGE OF CANDIDATE–STATES ELIGIBLE IN COMPLETE REFERENCES. GAPS ARE THE FIRST/LAST-STATE COMPUTED $(s_k - s_{k+1})/\max(1, |s_k|)$; THEY DESCRIBE THE STORED MODEL, NOT PHYSICAL SEPARATION ACCURACY OR A FULL GAP DISTRIBUTION.

Relation	Regime	k	Support (%)	First gap	Last gap
NIST	near	1	87.5	0.38	0.37
NIST	near	50	87.5	0.0024	0.0025
NIST	far	1	85.4	0.32	0.39
NIST	far	50	85.4	0.0084	0.012
CRAFTED	near	1	93.2	0.00054	4.6e-06
CRAFTED	near	50	93.2	0.0016	0.00079
CRAFTED	far	1	96.1	0.0025	0.004
CRAFTED	far	50	96.1	0.00055	0.0037

$k = 200$, both regimes lose to Incumbent. The far cell also loses to LeanCap, 1.168 [1.068,1.227], with Incumbent ratio 1.655 [1.344,2.279]; near LeanCap is mixed.

All six small-arrival CRAFTED cells in Table VII beat both controls in every block at 16 states. Under wide arrivals, only two Incumbent comparisons remain all-block wins and four are mixed; LeanCap has four wins and two mixed outcomes. Five wide-arrival comparisons with Roots–Only lose every block, with one mixed. NIST has further adverse cases: wide arrivals at $k = 50$, seeds 31002 and 31003, lose to Incumbent, while seeds 31001 and 31002 lose to LeanCap. The complete table retains the mixed and favorable alternatives. These matched-length trajectories probe the query regime, not a new chemistry population.

The shared kernel uses fixed positive-interval and direct-cap specializations. A separate 16-state implementation comparison favors specialized over generic Transport in all eight configurations, but both far NIST cells still lose to specialized LeanCap and far $k = 50$ also loses to Incumbent. The companion retains all 24 paired contrasts and their separate clocks. This tests a bundled implementation, not a new ranking mechanism or an isolated micro-optimization.

E. Numerical Verification and Reproduction

The preserved predecessor failed 16 ordered-ID cases on a five-pair dyadic Henry fixture at $P = 0.125$, $y = 1-2^{-40}$. The current kernel passes the unchanged 2,280-check suite, including those counterexamples. Expanded stresses cover 12,096 queries, 360 seeds, 2,156 threshold tests (294 unknown), 96,926 retained intervals, 59,136 transported intervals and 336 paired LeanCap queries, without an observed exported-enclosure miss. Ordinary regression covers another 1,024 queries and 16 seeds; full fixtures and failure outputs remain in the companion.

Across the original 32,256 timed queries, recorded ambiguous-threshold, cap-support-refinement and counted precision-fallback totals are zero. There is no unresolved-event counter, and direct near-support binary128 solves are not counted: these zeros do not describe all precision branches.

The original independent Decimal audit checks 288 measured outputs at 16 fixed states: eight size/k states at $n = 256/16010$, $k = 200$, near/far, first/last query; and eight arrival states across both relations, small/wide and first/last query for seed 31001, including $k = 1$ outputs. Support, cutoff and contender checks leave no ambiguity or enclosure miss. Another 480 outputs match saved independent references without new root scans.

Fresh extraction/compilation replays one original full-study block: 10,752 queries, 160 seeds and 304 lifecycles. IDs, scores, brackets, designated nonclock counters and binary hash match. Accounting and canonical reductions also pass the eight-stage audit. Replay clocks are separate. These finite checks establish neither all-domain floating-point soundness nor physical IAST accuracy.

For the practical study, independent Decimal scans at queries 1, 32 and 64 of each operating relation verify 432 measured outputs, with no observed enclosure miss. Another 288 near/far outputs match saved independent references. Fresh-source replay matches nonclock outputs for 6,144 queries, 48 seeds and 96 lifecycles, and the binary hash, on the same host. Accounting covers all 18,432 timed outputs. An incomplete 3-of-16 near/far endpoint-diagnostic export (cause unresolved) is preserved; a separate 16-query untimed completion matches the original three rows without altering timings. The initial failed audit and completed augmentation remain distinct records.

The original inputs, attribution, frozen kernel/protocols and full raw/canonical records are in `ICDE_C13_Science_Companion_v4.zip`. The additional six-policy studies and their preserved failed/completed audits are in `ICDE_C13_Repair1_Science_Companion.zip`, bound by accepted snapshot `C13_R5_REPAIR1_EVIDENCE_V1`. Their manifests bind source/data identities; replay and independent-check scopes are reported above.

VIII. DISCUSSION AND CONCLUSION

Partial-pressure ratios turn genuine root enclosures into conditional destination bounds. Eligible distinct-ID witnesses make those bounds useful before new integral evaluation. Matched paid comparisons, including on-demand warm solving, demonstrate the solved-root benefit on the original near/far CRAFTED streams. Fresh negative witnesses share the interface but do not consistently improve time.

One-query prefixes, small/high-k pools, wide arrivals and unfavorable acquisition states expose losses. Even the same operating grid gains or loses under different source states. These results concern binary finite-support ranking within the shared kernel; native external solver competitiveness and observed user-session prevalence remain unmeasured.

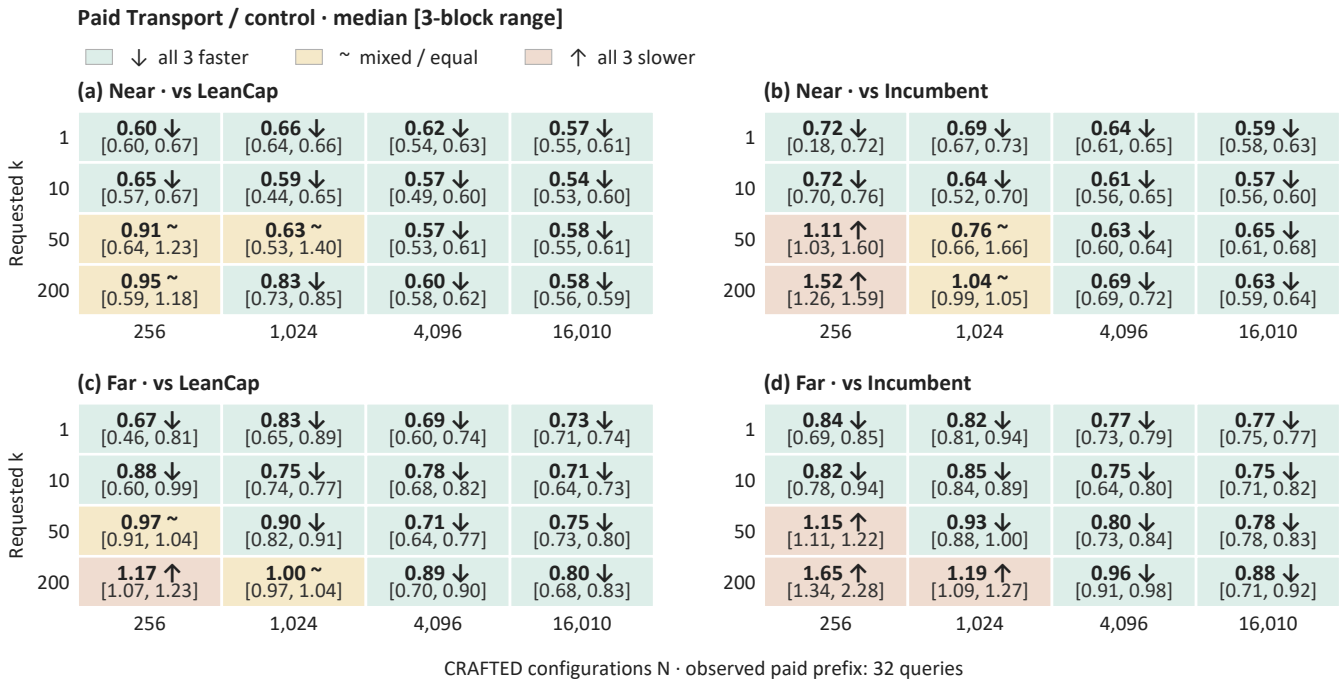


Fig. 4. Paid Transport/control ratios at 32 queries for all 32 CRAFTED size/k cells and fixed LeanCap/Incumbent controls: median [minimum, maximum] of three paired blocks. Teal ↓: all blocks faster; amber ∼: mixed/equal; copper ↑: all slower, using unrounded ratios. Nested pools use one fixed permutation.

TABLE VII

ALL 24 ARRIVAL CELLS AT 16 STATES: PAID TRANSPORT/CONTROL MEDIAN [MINIMUM, MAXIMUM] FROM THREE PAIRED BLOCKS. L: LEANCAP; I: INCUMBENT; R: ROOTS–ONLY. S1–S3 ARE SEPARATE STREAMS WITH SEEDS 31001–31003. TEAL ↓: ALL BLOCKS FASTER; AMBER ∼: MIXED/EQUAL; COPPER ↑: ALL SLOWER, CLASSIFIED BEFORE ROUNDING.

Relation	Arrivals	k	Seed	Ratio to L	Ratio to I	Ratio to R
NIST	small	1	S1	0.953 [0.781,1.030]$^{\sim}$	0.802 [0.601,0.943]$^{\downarrow}$	0.958 [0.947,0.994]$^{\downarrow}$
NIST	small	1	S2	0.962 [0.949,1.015]$^{\sim}$	0.853 [0.843,0.917]$^{\downarrow}$	0.985 [0.956,1.036]$^{\sim}$
NIST	small	1	S3	0.920 [0.826,1.002]$^{\sim}$	0.895 [0.876,0.960]$^{\downarrow}$	0.969 [0.795,0.988]$^{\downarrow}$
NIST	small	50	S1	1.066 [0.955,1.200]$^{\sim}$	1.098 [0.928,1.283]$^{\sim}$	1.299 [0.940,1.414]$^{\sim}$
NIST	small	50	S2	0.882 [0.839,0.889]$^{\downarrow}$	0.979 [0.964,0.989]$^{\downarrow}$	0.947 [0.943,0.974]$^{\downarrow}$
NIST	small	50	S3	1.017 [0.910,1.199]$^{\sim}$	1.168 [1.023,1.168]$^{\uparrow}$	1.118 [0.985,1.298]$^{\sim}$
NIST	wide	1	S1	0.963 [0.843,1.005]$^{\sim}$	0.944 [0.927,0.996]$^{\downarrow}$	1.020 [0.978,1.107]$^{\sim}$
NIST	wide	1	S2	1.109 [0.849,1.150]$^{\sim}$	0.956 [0.762,1.103]$^{\sim}$	1.000 [0.895,1.197]$^{\sim}$
NIST	wide	1	S3	0.982 [0.950,1.000]$^{\downarrow}$	0.874 [0.866,0.932]$^{\downarrow}$	0.938 [0.907,0.980]$^{\downarrow}$
NIST	wide	50	S1	1.127 [1.084,1.239]$^{\uparrow}$	1.196 [0.977,1.376]$^{\sim}$	1.052 [1.029,1.107]$^{\uparrow}$
NIST	wide	50	S2	1.068 [1.065,1.149]$^{\uparrow}$	1.215 [1.174,1.220]$^{\uparrow}$	1.059 [1.020,1.064]$^{\uparrow}$
NIST	wide	50	S3	0.925 [0.762,0.935]$^{\downarrow}$	1.154 [1.151,1.207]$^{\uparrow}$	0.953 [0.928,1.022]$^{\sim}$
CRAFTED	small	1	S1	0.720 [0.718,0.737]$^{\downarrow}$	0.759 [0.748,0.797]$^{\downarrow}$	0.965 [0.962,1.036]$^{\sim}$
CRAFTED	small	1	S2	0.723 [0.700,0.757]$^{\downarrow}$	0.758 [0.753,0.771]$^{\downarrow}$	1.005 [1.001,1.046]$^{\uparrow}$
CRAFTED	small	1	S3	0.690 [0.680,0.740]$^{\downarrow}$	0.738 [0.715,0.748]$^{\downarrow}$	1.033 [0.967,1.097]$^{\sim}$
CRAFTED	small	50	S1	0.707 [0.694,0.741]$^{\downarrow}$	0.753 [0.722,0.770]$^{\downarrow}$	0.994 [0.990,1.035]$^{\sim}$
CRAFTED	small	50	S2	0.700 [0.695,0.701]$^{\downarrow}$	0.743 [0.724,0.762]$^{\downarrow}$	0.989 [0.941,0.992]$^{\downarrow}$
CRAFTED	small	50	S3	0.689 [0.644,0.700]$^{\downarrow}$	0.697 [0.688,0.710]$^{\downarrow}$	0.923 [0.877,0.989]$^{\downarrow}$
CRAFTED	wide	1	S1	0.840 [0.809,0.984]$^{\downarrow}$	0.908 [0.811,0.946]$^{\downarrow}$	1.075 [1.007,1.090]$^{\uparrow}$
CRAFTED	wide	1	S2	0.955 [0.946,0.968]$^{\downarrow}$	1.001 [0.998,1.012]$^{\sim}$	1.042 [0.962,1.062]$^{\sim}$
CRAFTED	wide	1	S3	0.898 [0.892,0.901]$^{\downarrow}$	0.923 [0.913,0.969]$^{\downarrow}$	1.019 [1.014,1.050]$^{\uparrow}$
CRAFTED	wide	50	S1	0.966 [0.948,0.969]$^{\downarrow}$	1.016 [0.973,1.040]$^{\sim}$	1.038 [1.022,1.048]$^{\uparrow}$
CRAFTED	wide	50	S2	0.991 [0.976,1.001]$^{\sim}$	1.024 [0.999,1.035]$^{\sim}$	1.076 [1.068,1.130]$^{\uparrow}$
CRAFTED	wide	50	S3	0.968 [0.944,1.006]$^{\sim}$	1.000 [0.960,1.029]$^{\sim}$	1.084 [1.081,1.137]$^{\uparrow}$

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